

Ming Gong, Ph.D.

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Summary

Seasoned **engineering and applied-sciences leader** with 13+ years building consumer-scale **search, ranking, recommendation, and Generative AI** products reaching hundreds of millions of users. Currently **Head of ML Engineering, Knowledge AI** at Atlassian — single point of accountability for an extended org of **130+** across Search & Ranking, Knowledge Graph, Generative Answers, and AI Agents. Previously **Group Senior Principal Applied Sciences Manager at Microsoft** (11 years), built and led 40+ engineers/scientists shipping Bing Question-Answering, Knowledge Card, Recommendation, and Copilot Search from zero to **100+ languages and 230+ markets**.

Deep expertise spanning **ranking & relevance modeling, recommendation systems, search infrastructure, query understanding, Generative AI (LLM post-training, SLM fine-tuning, agentic frameworks)**, and large-scale ML serving. Uniquely multi-dimensional: **applied sciences & full-stack engineering, AI products & research** (50+ publications, 11K+ citations, 20+ patents), **consumer & enterprise products**.

Selected Highlights

- **Ranking & recommendation at consumer scale:** Shipped ranking and recommendation systems across Bing and Atlassian — Knowledge Card (**65M DAU, +3.0% WPSBS**), Segment Entity Recommendation (**beat Google on relevance**), Jira Search (**3.4× MAU growth**). Built end-to-end ranking stacks from feature engineering and DNN/transformer modeling to online serving.
- **Recommendation infrastructure & signals:** Built entity recommendation, collaboration graph, entity linking (**40% → 90%+ precision**), and content signal platforms adopted across multiple product teams. Co-owned NRT inference pipeline with **8 platform teams across time zones**; designed unified NLU platform powering **95%+ of Bing answers**.
- **Generative AI:** Led Copilot Search (multi-turn AI search vertical) — **shipped one month before Google's AI Mode**. Built Glow SLM framework replacing GPT-4-class models (**20–30 s → ~5 ms latency**). Generative Answers covering **20% of Bing global queries, +2.0% WPSBS**. At Atlassian, drove AI MAU from **100K → 1.2M** via generative search experiences.
- **Global & multilingual:** Pioneered cross-lingual scaling methodology for search & QnA across **100+ languages, 230+ markets** — universal model on **~13% of per-language GPU footprint; 2× Microsoft MLADS Distinguished Contribution Awards** (first-ever back-to-back).
- **Ship velocity & 0→1 products:** Jira Search MVP in **4 months**, full launch within 12 months; Enterprise GenAI Search POC in **<1 month**; Copilot Search from concept to ship in **~12 months**. First BERT model deployed in Bing **within 1 month of open-source**.
- **Org building:** Single POC for **extended org of 130+** at Atlassian; previously built 40+ team at Microsoft from scratch across two sites. Recruited principal-level managers and tech leads; established greenfield sites.
- **Research:** Best Paper Award, IJCAI '24; 50+ publications, 11K+ citations, 20+ patents. Co-authored CodeBERT (3,000+ citations), Unicoder, and pioneering work in dense retrieval and knowledge representation.

Experience

Atlassian (US)

Feb 2025 – Present

Head of Machine Learning Engineering — Knowledge AI

Bellevue, WA

- Single point of accountability for the Knowledge AI charter, leading an **extended org of 130+** — 60+ direct reports (70% scientists, 30% engineers; full-stack and globally distributed) plus dotted-line partner teams across Jira Search & Agent, Knowledge Fundamentals, and Generative AI Search. Grew the team from 30 to 60+ in one year, recruiting **2 Principal-level Managers and 3 Principal-level Tech Leads**; established new teams in **Canada and Singapore**.
- **Search ranking & recommendation:** Shipped Jira Search MVP within 4 months; rebuilt ranking stack and search UX, driving **3.4× Search MAU growth (2.5M → 8.5M)**, helping AI Org exceed Search MAU L1 OKR. Formed a **V-Team growing from ~30 to 115 contributors across 8 partner orgs** with a Coalition Council for cross-org alignment.
- **Knowledge signals & entity recommendation:** Built core knowledge fundamentals — collaboration graph, entity linking, entity recommendation — now adopted by multiple product teams for personalized features. Led **entity linking turnaround from 40% to 90%+ precision**; invested 2 months in hands-on diagnosis, protected team stability through parallel senior hiring, then redirected technical approach.
- **Generative AI products:** Generative Answers in search and AI Definitions drove **AI MAU from 100K to 1.2M**. Initiated and led **Enterprise Generative AI Search** from zero to GA — POC in <1 month, MVP shipped early 2026. Built **Lumina** generative UI framework — **20% engagement uplift, 30% SBS lift** vs. competitor AI experiences; adopted as Atlassian's standard generative presentation framework.

- **Agentic AI:** Launched **Jira Agents in Rovo Chat** within 3 months, becoming a primary driver of Rovo adoption. Code-driven agent architecture **beat Manus and GenSpark on the GAIA benchmark**.

Microsoft — Search Technology Center Asia (Bing, Edge, Copilot)

Aug 2013 – Dec 2024

Group Senior Principal Applied Sciences Manager (2019–2024) · Senior Dev Manager (2018–2019) · Senior Tech Lead (2016–2018) · SWE II (2013–2015) *Beijing / Suzhou*

- **Built and led an org of 40+ engineers and applied scientists from scratch** (70% scientists, 30% engineers; **5 managers**) across two sites, spanning Bing Search, Copilot, QA, Knowledge Graph, Recommendation, and Edge browser AI.
- **Copilot Search — AI search vertical (2024–2025):** delivered multi-turn AI search experience unifying GenQnA, GenSERP, and Magazine; built scalable LLM generation pipelines and task-specific SLMs (incl. R1-distilled). **Shipped April 2025 — one month before Google’s AI Mode.**
- **Knowledge Card — ranking & recommendation for 150M entities (2021–2023):** multi-modal entity aggregation across web sources with template + MRC DL extraction and LLM generation. **60% Bing query traffic, 65M DAU, +3.0% WPSBS;** certified as a Bing “WoW” feature (<10 out of 100+ features).
- **Segment Entity Search & Recommendation (2021–2024):** shipped recipe, job, and movie recommendation answers — **beat Google on relevance and WPSBS** on recipe and job; closed WPSBS gap with Google on movie recommendation. End-to-end ranking pipeline from content signals to relevance modeling and delivery.
- **Generative Answers on Bing SERP (2023–2024):** **20% query coverage on Bing global markets, +2.0% WPSBS** via streaming generative answers and click-through to Bing Chat.
- **Generative SERP (2023–2024):** reinvented Bing SERP as “AI Masterpieces” — proposed tier-wise hallucination control framework; built **Glow**, a distilled SLM framework replacing GPT-4-class models — first-token latency **20–30 s** → **~5 ms**; scaled to 5–10% query coverage.
- **Question-Answering: zero to global, ML → DL → LLM (2016–2024):** built Bing QnA from the ground up and led its evolution through every major AI paradigm shift. Scaled from English-only to **100+ languages and 230+ markets**. Shipped 10+ DNN releases that replaced legacy GBDT ranker (400+ hand-crafted features) and won WPSBS vs. Google. Authored novel BERT compression — first BERT model deployed in Bing (**within 1 month of open-source, zero additional GPU cost**).
- **Query-Dependent Passage Extraction (2019–2022):** replaced industry-standard sliding-window passage indexing with runtime query-dependent extraction via GNN + multi-teacher distillation. **Held #1 on Google Natural Questions benchmark for ~1 year;** architecture adopted across Bing QnA, Knowledge, People Also Ask.
- **Edge AI — Underside right-rail panel & Copilot (2022–2024):** built AI-powered right-rail panel (summary, topics, QnA, related search, content feeds). **Edge engaged DAU +10%, Search DAU +1.2M.** Long-document summarization for Edge/Windows Copilot — 1000+ pages at **95%+ accuracy**.
- **NLU Platform “Orca” (2020–2025):** unified query understanding pipeline powering answer triggering for **95%+ of Bing answers;** 60% of onboarding answers closed triggering gap with Google.

Platforms & Internal Tooling Built

- **Lumina** (Atlassian, 2025–2026) — generative UI framework for Enterprise GenAI Search; **20% engagement uplift, 30% SBS lift;** adopted as Atlassian’s standard generative presentation framework across all GenAI products.
- **X-SuperAgent** (Atlassian, 2025–2026) — Atlassian’s first OpenClaw-compliant agent framework for AI-native team operation; **40%+ developer-productivity uplift.**
- **Glow** (Microsoft, 2023–2024) — distilled SLM serving framework replacing GPT-4-class models for production; first-token latency **20–30 s** → **~5 ms**; V2 scaled to 5–10% Bing query coverage.
- **Orca** (Microsoft, 2020–2025) — unified NLU platform (domain, intent, slots, topics) powering answer triggering for **95%+ of Bing answers.**
- **NeuronBlocks** (Microsoft, 2019, open source) — “Build NLP DNN models like playing Lego.” Published at EMNLP/IJCNLP 2019.

People & Org Development

- **Founder & Head, Microsoft AI School — Asia-Pacific R&D Group (2017–2024).** Led a **50+ operation v-team; 10K+ participants, 2K+ honored graduates;** 8 years running. Many graduates grew into AI talents and leaders across the Microsoft APRD org.
- **Invited instructor, LinkedIn Global Academy for Chinese Programmers (2022–2023):** “Intelligent QA Systems for Search Engines” — **50M impressions, 98% satisfaction.**
- **Invited speaker, InfoQ AIcon Global AI Conference (2021):** represented Microsoft China on globalization of Bing QA.

Education

Ph.D., Graphics and Visual Computing

2008 – 2013

Institute of Computing Technology, Chinese Academy of Sciences

Selected Awards

- **Best Paper Award**, IJCAI '24 AI4Research — “*Step-Back Profiling: Distilling User History for Personalized Scientific Writing.*”
- **2021 Fall Microsoft MLADS Distinguished Contribution Award** (2 / 250+). Broke conference history to receive the award twice consecutively.
- **2021 Spring Microsoft MLADS Distinguished Contribution Award** (2 / 300+).
- 2023 Microsoft Global Hackathon — Edge Theatre: **Best Overall Hack**; Greater China Region: **Most Popular Hack**.
- 2023 Microsoft Global Hackathon — DeKG (decentralized knowledge ecosystem): **Best Innovativeness & Originality Hack**.
- Microsoft AI Core Q3 FY18 **Greatness Award** (2018).

Selected Publications (50+ papers, 11,000+ citations — full list on scholar profile)

▷ Recommendation & Personalization

- Ning Wu, **Ming Gong**, Linjun Shou, Jian Pei, Daxin Jiang. “RUEL: Retrieval-Augmented User Representation with Edge Browser Logs for Sequential Recommendation.” *CIKM* 2023.
- Hanshuang Tong, Jun Li, Ning Wu, **Ming Gong**, et al. “Plutos: Towards Interpretable Stock Movement Prediction with Financial Large Language Model.” *arXiv* 2024.

▷ Dense Retrieval & RAG

- Shengyao Zhuang, Linjun Shou, Jian Pei, **Ming Gong**, et al. “Typos-aware Bottlenecked Pre-Training for Robust Dense Retrieval.” *SIGIR* 2023.
- Houxing Ren, Linjun Shou, Jian Pei, Ning Wu, **Ming Gong**, Daxin Jiang. “Lexicon-Enhanced Self-Supervised Training for Multilingual Dense Retrieval.” *EMNLP* 2023.
- Shunyu Zhang, Yaobo Liang, **Ming Gong**, Daxin Jiang, Nan Duan. “Modeling Sequential Sentence Relation to Improve Cross-lingual Dense Retrieval.” *ICLR* 2023.
- Shunyu Zhang, Yaobo Liang, **Ming Gong**, Daxin Jiang, Nan Duan. “Multi-View Document Representation Learning for Open-Domain Dense Retrieval.” *ACL* 2022.
- Linjun Shou, S. Bo, Fei-Xiang Cheng, **Ming Gong**, et al. “Mining Implicit Relevance Feedback from User Behavior for Web Question Answering.” *KDD* 2020.

▷ LLM, Agentic AI & Multimodal Pre-training

- Yaobo Liang, Chenfei Wu, Ting Song, et al., **Ming Gong**, Nan Duan. “TaskMatrix.AI: Completing Tasks by Connecting Foundation Models with Millions of APIs.” *Intelligent Computing* 2023.
- Gen Li, Nan Duan, Yuejian Fang, **Ming Gong**, Daxin Jiang. “Unicoder-VL: A Universal Encoder for Vision and Language by Cross-modal Pre-training.” *AAAI* 2019.
- Nuo Chen, Ning Wu, Shining Liang, **Ming Gong**, et al. “Is Bigger and Deeper Always Better? Probing LLaMA Across Scales and Layers.” *arXiv* 2023.
- Ning Wu, **Ming Gong**, Linjun Shou, et al. “Large Language Models are Diverse Role-Players for Summarization Evaluation.” *NLPCC* 2023.

▷ Model Training, Compression & Distillation

- **Ming Gong**, Linjun Shou, Wutao Lin, et al. “NeuronBlocks — Building Your NLP DNN Models Like Playing Lego.” *EMNLP/IJCNLP* 2019.
- Ze Yang, Linjun Shou, **Ming Gong**, Wutao Lin, Daxin Jiang. “Model Compression with Two-stage Multi-teacher Knowledge Distillation for Web Question Answering System.” *WSDM* 2020.
- Fei Yuan, Linjun Shou, Jian Pei, Wutao Lin, **Ming Gong**, Daxin Jiang. “Reinforced Multi-Teacher Selection for Knowledge Distillation.” *AAAI* 2021.

▷ Code Intelligence

- Zhangyin Feng, et al., **Ming Gong**, Linjun Shou, et al. “CodeBERT: A Pre-Trained Model for Programming and Natural Languages.” *EMNLP (Findings)* 2020. (3,000+ citations)

▷ Cross-Lingual & Multilingual Modeling

- H. Huang, Yaobo Liang, N. Duan, **Ming Gong**, et al. “Unicoder: A Universal Language Encoder by Pre-training with Multiple Cross-lingual Tasks.” *EMNLP/IJCNLP* 2019.
- Nuo Chen, Linjun Shou, **Ming Gong**, Jian Pei. “From Good to Best: Two-Stage Training for Cross-lingual Machine Reading Comprehension.” *AAAI* 2022.

- Shining Liang, **Ming Gong**, Jian Pei, Linjun Shou, Daxin Jiang. “Reinforced Iterative Knowledge Distillation for Cross-Lingual Named Entity Recognition.” *KDD* 2021.
- ▷ **Knowledge & Entity Understanding**
- Zilin Xiao, **Ming Gong**, Jie Wu, et al. “Instructed Language Models with Retrievers Are Powerful Entity Linkers.” *EMNLP* 2023.
- Zilin Xiao, Linjun Shou, Xingyao Zhang, Jie Wu, **Ming Gong**, et al. “Coherent Entity Disambiguation via Modeling Topic and Categorical Dependency.” *Findings of EMNLP* 2023.

Selected Patents (20+ granted; selection below)

- Personalized Multilingual Content Recommendation based on Robust Feature Network. 2023.
- Sequential Recommendation based on Cross-Domain Behavior Data. 2023.
- Sequential Recommendation Based on Dual Contrastive Interest Learning. 2022.
- Content Recommendation Based on Graph Enhanced Collaborative Filtering. 2022.
- Knowledge-enhanced Content-Aware Collaborative Filtering for Recommendation. 2021.
- Hierarchical Representation Learning of User Interest. 2021.
- Model Generation based Model Compression. 2019.
- Model Selection Learning for Knowledge Distillation. 2020.
- Contrastive Learning for Question Answering. 2020.
- Providing QA Training Data and Training a QA Model based on Implicit Relevance Feedbacks. 2020.
- Lexicon-Enhanced Self-Supervised Training for Multilingual Dense Retrieval. 2022.
- Graph based Fact Clustering for Diversity and Freshness. 2021.
- Temporal Co-Contrastive Learning-Based Node Representation Generation. 2022.